

# Software Engineering Project Proposal: StudyPlanner Pro (Web Edition)

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February 16, 2026

**Statement of Integrity:** I have followed the University Code of Conduct and Student Honor Code. This work was completed entirely by me and my groupmates. I have not used any unauthorized internet help, nor am I aware of any other person violating this code.

## 1 Analyzing the Problem

### 1.1 The User: Sarah's Story

The primary user for this project is "Sarah," a second-year university student majoring in Business. Sarah is a hardworking student, but she is currently overwhelmed by her workload. This semester, she is taking five difficult courses, each with different professors, credit values, and grading scales. Like many students, Sarah has a busy life that includes a part-time job and social activities with her friends.

Sarah currently tries to stay organized using a paper-based system. She has a physical notebook where she writes down her assignments and a calendar on her wall. However, she finds that this manual system is failing her. She often forgets to check her notebook when she is at the library, or she loses track of a small assignment mentioned in a syllabus weeks ago. Because her information is scattered in different places, she feels a constant sense of "academic stress" and anxiety.

### 1.2 Specific Information Management Problems

The first major problem is the difficulty of tracking academic progress. Sarah wants to know her current Grade Point Average (GPA) to see if she is meeting her goals. Currently, she has to find a calculator, look up the credit weight for each class, and manually do the math. This is time-consuming and prone to errors. If she makes one mistake in her calculation, she might have a false sense of security or unnecessary panic about her grades.

The second problem is time management and prioritization. Sarah often has ten different "unfinished tasks" on her list. Some are due tomorrow, and some are due in three weeks. On paper, they all look the same. She has no way to see a "countdown" that shows her exactly how much time is left. Without this visual pressure, she often procrastinates on big projects until they become emergencies. Research shows that about 90% of college students procrastinate, and 47% feel their high school did not teach them the organizational skills needed for university life.

### 1.3 Current Solutions and Their Failures

Currently, Sarah uses a mix of three things: a paper notebook, phone alarms, and her memory.

1. **The Paper Notebook:** It is messy and difficult to search. She cannot "filter" her tasks to see only what is due today. It also cannot send her notifications or show a live countdown.
2. **Phone Alarms:** These are noisy and annoying. They tell her *when* to wake up, but they don't help her plan her study session or show her her GPA.
3. **Memory:** This is the most dangerous "system." Sarah often thinks she will remember a deadline, but during busy weeks, small tasks "slip through the cracks."

Sarah's current solutions are "static." They do not change as time moves. In contrast, a software solution is "dynamic." It can update her GPA instantly when a new grade is entered and show a ticking clock for her deadlines.

### 1.4 Why a Web-Based Software Helps

A web-based application (Web GUI) is the perfect solution for Sarah. Unlike a basic computer program that only stays on one laptop, a Web GUI allows Sarah to log in from the computer lab, her laptop at home, or even her phone.

By moving to a digital system, Sarah can benefit from "automation." The computer can handle all the math for her GPA and total credits. It can also use a "Recommendation Engine" to look at all her deadlines and suggest the most important task to work on today. This removes the "decision fatigue" of trying to figure out what to do first. Furthermore, adding a "Daily Luck" feature adds a layer of fun. By using her zodiac sign to give a "Study Fortune," the software becomes a friendly companion rather than just a cold database.

### 1.5 Detailed Inputs and Outputs

To solve these problems, the system will manage the following specific data:

## System Inputs:

- **Course Information:**

- Course Name (e.g., "Math 101")
- Professor Name (e.g., "Dr. Smith")
- Credits (e.g., 3.0 or 4.0)
- Grade (The current letter or percentage grade Sarah has earned)

- **Task Information:**

- Task Name (e.g., "Final Essay")
- Deadline (The specific date and time the task is due)

## System Outputs:

- **Study Statistics:** A summary showing the calculated GPA, Total Earned Credits, the number of Unfinished Tasks, and a list of Completed Tasks.
- **Today's Recommended Task:** A single task highlighted by the system as the most urgent thing to do today.
- **Task Countdown:** A visual timer for each task (e.g., "3 days, 2 hours remaining").
- **Daily Study Luck (Optional):** A fun message based on Sarah's zodiac sign to encourage her.

## 2 Criteria for Success

The requirements below act as a "contract" for the project. If the finished Web App does these things, it is a success.

### 2.1 Functional Requirements (What the system must do)

1. **Account Management:** The system shall allow the user to create an account and log in securely so their data is private.
2. **Course Entry:** The user must be able to add, edit, or delete courses. Each course must include the name, professor, credits, and grade.
3. **Task Tracking:** The user must be able to add tasks with a name and a specific deadline.
4. **GPA Calculation:** The system shall automatically calculate the GPA. It will use a weighted formula:  $GPA = \frac{\sum(GradeValue \times Credits)}{\sum Credits}$ .
5. **Credit Summation:** The system must display the total number of credits entered in the system.
6. **Task Status:** The user must be able to mark a task as "Finished." The system will then move it from the "Unfinished" list to the "Completed" list.
7. **Live Countdown:** For every unfinished task, the system must display a real-time countdown showing how much time is left.
8. **Smart Recommendation:** The system shall analyze all unfinished tasks and display one "Recommended Task" on the main dashboard. This will usually be the task with the closest deadline.

### 2.2 User Interface Requirements (Web GUI)

1. **Web Dashboard:** The main page should be a "Dashboard" layout. This means the most important info (GPA, Today's Task, and Countdown) is visible immediately without clicking anything.
2. **Navigation Bar:** The system shall have a clear navigation menu to move between "Courses," "Tasks," and "Statistics."
3. **Visual Hierarchy: (Optional)** Urgent tasks (deadlines less than 24 hours away) should be highlighted to catch the user's attention.
4. **Responsive Design:** The layout must work on both a computer screen and a mobile phone browser.
5. **Clean Aesthetic:** The design should be simple and professional, not "messy" like a paper notebook.

## 2.3 Optional Features (The "Extra" Value)

1. **Daily Study Luck:** The system will include an optional "Fortune" module. After the user enters their birthday, the system will identify their zodiac sign. Every day, it will provide a simple, fun message (e.g., "Aries: Today is a lucky day for Math! Focus and you will succeed.").
2. **Progress Bars:** For each course, the system can show a visual bar (0% to 100%) based on how many tasks for that course are finished.

### 3 Prototype

StudyPlanner Pro will be a modern Web Application. Below is a prototype.

```
=====
      STUDYPLANNER PRO - MAIN MENU
=====
1. View My Stats (GPA & Credits)
2. Add a New Course
3. Add a New Task
4. View Today's Recommendation
5. Exit

Select an option: 1

--- MY STUDY STATS ---
Current GPA: 3.8
Total Credits: 12
Completed Tasks: 5
Unfinished Tasks: 2

[Press Enter to return to Menu]

Select an option: 4

--- TODAY'S RECOMMENDATION ---
Based on your deadlines, you should work on:
>> "Math Quiz" for Prof. Smith's class.
COUNTDOWN: 0 Days, 22 Hours remaining!
[Finish it now to keep your 3.8 GPA!]

Select an option: 2

Enter Course Name: Data Structures
Enter Professor: Dr. Zhang
Enter Credits: 4
Enter Current Grade: 4.0
Course Added Successfully!
=====
```